

36.2 Cumulative logit models

Suppose the c categories are ordered: none, mild, severe; strongly disagree through strongly agree; extremely liberal through extremely conservative. The baseline-category model of [Section 36.1](#) fits such data perfectly well — it fits *any* categorical response — but throws the ordering away. It spends $(c - 1)p$ coefficients where a model that knows the order can often spend p , and its fit is unchanged if the category labels are shuffled. That last property is the diagnosis: a likelihood invariant to permuting the categories cannot be using the order.

36.2.1 Cutting the scale

The device that uses the ordering is to *cut*: for each r between 1 and $c - 1$, collapse the response into the binary event $\{Y_i \leq r\}$ and model its probability. Write

$$\gamma_{ir} = P(Y_i \leq r \mid \mathbf{x}_{(i)}) = \pi_{i1} + \cdots + \pi_{ir}, \quad r = 1, \dots, c - 1, \quad (36.2.1)$$

for the **cumulative probabilities**, with $\gamma_{i0} = 0$ and $\gamma_{ic} = 1$. The category probabilities are recovered by differencing, $\pi_{ir} = \gamma_{ir} - \gamma_{i,r-1}$, so modelling the $c - 1$ cumulative probabilities is modelling the whole distribution.

Definition 36.2.1. Cumulative link model

Let G be a continuous, strictly increasing distribution function on $\mathbb{R}$ and let $\mathbf{x}_{(i)}$ be a vector of p regressors *without* an intercept. The **cumulative link model** is

$$G^{-1}(\gamma_{ir}) = \theta_r - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r, \quad r = 1, \dots, c - 1, \quad (36.2.2)$$

with **cutpoints** $\theta_1, \dots, \theta_{c-1}$ and slope vectors $\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_{c-1}$. With $G = \Lambda$, the standard logistic distribution function $\Lambda(z) = e^z / (1 + e^z)$, [Eq. 36.2.2](#) is the **cumulative logit model**

$$\log \frac{P(Y_i \leq r \mid \mathbf{x}_{(i)})}{P(Y_i > r \mid \mathbf{x}_{(i)})} = \theta_r - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r;$$

with $G = \Phi$ it is the **cumulative probit model**, and with $G(z) = 1 - \exp(-e^z)$ the **cumulative complementary log-log model**. The special case $\boldsymbol{\beta}_1 = \cdots = \boldsymbol{\beta}_{c-1} = \boldsymbol{\beta}$ is the **proportional odds model** when $G = \Lambda$, and is the subject of [Section 36.3](#).

Two conventions deserve comment. The intercept is not in $\mathbf{x}_{(i)}$ because the cutpoints already play that role: shifting every θ_r and a hypothetical intercept by the same constant would change nothing, so the two cannot both be free. And the minus sign before $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r$ is worth the confusion it causes once, because it makes a *positive* coefficient mean “larger values of this regressor push the response towards the higher categories”. The cutpoints are unconstrained parameters only if the model is coherent, which is the one real difficulty of this family.

36.2.2 The ordering constraint

Proposition 36.2.1. Cutpoints must stay in order

Eq. 36.2.2 defines a probability distribution over the c categories at $\mathbf{x}_{(i)}$ if and only if

$$\theta_1 - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_1 < \theta_2 - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_2 < \cdots < \theta_{c-1} - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_{c-1}. \quad (36.2.3)$$

If the slopes are common, $\boldsymbol{\beta}_r = \boldsymbol{\beta}$ for all r , this reduces to $\theta_1 < \cdots < \theta_{c-1}$, a constraint on the parameters alone. If the slopes differ, the constraint involves $\mathbf{x}_{(i)}$, and for any two cuts with $\boldsymbol{\beta}_r \neq \boldsymbol{\beta}_{r+1}$ there are values of $\mathbf{x}$ at which it fails.

Proof. The category probabilities are $\pi_{ir} = \gamma_{ir} - \gamma_{i,r-1}$, so they are all positive exactly when $\gamma_{i1} < \cdots < \gamma_{i,c-1}$, and they automatically sum to one. Since G is strictly increasing, this is Eq. 36.2.3. With common slopes the terms $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ cancel from every inequality. If $\mathbf{d} = \boldsymbol{\beta}_{r+1} - \boldsymbol{\beta}_r \neq \mathbf{0}$, the r th inequality reads $\mathbf{x}^\top \mathbf{d} < \theta_{r+1} - \theta_r$, a half-space; any $\mathbf{x}$ far enough in the direction $\mathbf{d}$ violates it. ■

Warning (What the constraint costs). The proportional odds model is safe: the constraint holds for every $\mathbf{x}$ at once, and by Exercise (B1) it is satisfied automatically at the maximum likelihood estimate. A model with cut-specific slopes is *not* a model over the whole of $\mathbb{R}^p$, only on the region where Eq. 36.2.3 holds; one fits it, checks that the fitted probabilities are positive wherever one intends to predict, and reports the check. For a concrete failure take $c = 3$, one regressor, $\theta_1 = 0$, $\beta_1 = 1$, $\theta_2 = 1$, $\beta_2 = 2$: at $x = 0$ the linear predictors are 0 and 1, in order; at $x = 2$ they are -2 and -3 , out of order, and the fitted probability of the middle category is negative.

36.2.3 Fitting

The cumulative link model is not a generalized linear model in the strict sense of Definition 34.2.1: the response is a vector and the parameters are shared across its $c - 1$ coordinates in a way no single link expresses. The likelihood machinery survives intact, because the response is still multinomial and the mean is still a smooth function of the parameters.

Proposition 36.2.2. Score and expected information for a smooth mean

Let $\mathbf{Y}_i \sim \text{Mult}(m_i, \boldsymbol{\pi}_i)$ independently, $i = 1, \dots, n$, with $\boldsymbol{\pi}_i = \boldsymbol{\pi}_i(\boldsymbol{\psi})$ a smooth function of a parameter $\boldsymbol{\psi} \in \mathbb{R}^q$ and every $\pi_{ir} > 0$. Write $\tilde{\boldsymbol{\pi}}_i$ for the first $c - 1$ probabilities, $\mathbf{D}_i = \partial \tilde{\boldsymbol{\pi}}_i / \partial \boldsymbol{\psi}^\top$ and $\mathbf{V}_i = \text{diag}(\tilde{\boldsymbol{\pi}}_i) - \tilde{\boldsymbol{\pi}}_i \tilde{\boldsymbol{\pi}}_i^\top$. Then the score and the expected information are

$$\mathbf{u}(\boldsymbol{\psi}) = \sum_{i=1}^n \mathbf{D}_i^\top \mathbf{V}_i^{-1} (\tilde{\mathbf{y}}_i - m_i \tilde{\boldsymbol{\pi}}_i), \quad \mathbf{J}(\boldsymbol{\psi}) = \sum_{i=1}^n m_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{D}_i. \quad (36.2.4)$$

Proof. The inverse of $\mathbf{V}_i$ is $\text{diag}(1/\tilde{\boldsymbol{\pi}}_i) + \pi_{ic}^{-1} \mathbf{1}\mathbf{1}^\top$, as multiplying out and using $\mathbf{1}^\top \tilde{\boldsymbol{\pi}}_i = 1 - \pi_{ic}$ shows. Also $\partial \ell / \partial \boldsymbol{\psi} = \sum_i \sum_{r \leq c} (y_{ir} / \pi_{ir}) \partial \pi_{ir} / \partial \boldsymbol{\psi}$. Fix i , drop it from the notation, and

substitute the inverse:

$$\mathbf{D}^\top \mathbf{V}^{-1}(\tilde{\mathbf{y}} - m\tilde{\boldsymbol{\pi}}) = \sum_{r < c} \frac{\partial \pi_r}{\partial \boldsymbol{\psi}} \frac{y_r - m\pi_r}{\pi_r} + \left(\sum_{r < c} \frac{\partial \pi_r}{\partial \boldsymbol{\psi}} \right) \frac{\sum_{s < c} (y_s - m\pi_s)}{\pi_c}.$$

Because the probabilities sum to one, $\sum_{r < c} \partial \pi_r / \partial \boldsymbol{\psi} = -\partial \pi_c / \partial \boldsymbol{\psi}$, and because the counts sum to m , $\sum_{s < c} (y_s - m\pi_s) = -(y_c - m\pi_c)$. The second term is therefore $(\partial \pi_c / \partial \boldsymbol{\psi})(y_c - m\pi_c) / \pi_c$, and the two terms combine into $\sum_{r < c} (\partial \pi_r / \partial \boldsymbol{\psi})(y_r - m\pi_r) / \pi_r$, which equals $\sum_{r < c} (y_r / \pi_r) \partial \pi_r / \partial \boldsymbol{\psi}$ since $\sum_r \partial \pi_r / \partial \boldsymbol{\psi} = \mathbf{0}$. Summing over i gives the score. For the information, $\text{Cov}(\tilde{\mathbf{Y}}_i) = m_i \mathbf{V}_i$ by Lemma 36.1.1, and the summands of the score are independent with mean zero, so $\text{Cov}(\mathbf{u}) = \sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} (m_i \mathbf{V}_i) \mathbf{V}_i^{-1} \mathbf{D}_i = \sum_i m_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{D}_i$. ■

For Eq. 36.2.2 the derivatives are immediate: writing $\eta_{ir} = \theta_r - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r$ and $g = G'$ for the density,

$$\frac{\partial \gamma_{ir}}{\partial \theta_s} = g(\eta_{ir}) \delta_{rs}, \quad \frac{\partial \gamma_{ir}}{\partial \boldsymbol{\beta}_s} = -g(\eta_{ir}) \mathbf{x}_{(i)} \delta_{rs},$$

and $\mathbf{D}_i$ is obtained by differencing these rows. Fisher scoring $\boldsymbol{\psi} \leftarrow \boldsymbol{\psi} + \mathbf{J}^{-1} \mathbf{u}$ converges quickly from ordered starting cutpoints and zero slopes. Because Eq. 36.2.4 has the form “derivative of the mean, weighted by the inverse covariance, times the residual”, each step is again a weighted least squares fit — the block-weighted version of Theorem 34.4.1 and Proposition 20.6.1 — with $\mathbf{V}_i$ in the place of scalar weights.

Remark (Why the fitting is well behaved). Suppose the link density $g = G'$ is log-concave, as it is for the logistic, the normal and the extreme-value distribution behind the complementary log-log link. Then

$$G(a) - G(b) = \int g(u) \mathbf{1}\{b < u \leq a\} du$$

integrates over u a function of (u, a, b) that is log-concave on $\mathbb{R}^3$, being a log-concave function of u times the indicator of a convex set; a marginal of a log-concave function is log-concave, so $\log\{G(a) - G(b)\}$ is concave on $\{b < a\}$. The observation in category r contributes $\log\{G(\eta_{ir}) - G(\eta_{i,r-1})\}$ with η_{ir} affine in the parameters, so the log-likelihood is concave on the region where Eq. 36.2.3 holds at the observed $\mathbf{x}_{(i)}$ — an intersection of half-spaces, hence convex. That is Pratt (1981) and Burridge (1981). Concavity excludes local maxima that are not global, which is what makes scoring from any ordered start dependable; by itself it gives neither existence nor uniqueness, and with cut-specific slopes the region it holds on is cut out by the data.

Remark (Asymptotics, and what they rest on). This is not the canonical-link generalized linear model Theorem 34.5.1 is stated for, so that theorem cannot be quoted here. What holds is the standard theory of maximum likelihood in a smooth parametric model: suppose the model is correctly specified and identifiable, c and p are fixed, $\boldsymbol{\psi}^0$ is interior to the region where Eq. 36.2.3 holds, G is twice continuously differentiable with $g > 0$, the regressors are bounded, and $n^{-1} \mathbf{J}(\boldsymbol{\psi}) \rightarrow \mathbf{J}_\infty$ positive definite uniformly near $\boldsymbol{\psi}^0$. Then with probability tending to one the likelihood equations have a consistent solution, and $\sqrt{n}(\hat{\boldsymbol{\psi}} - \boldsymbol{\psi}^0)$ converges in distribution to $N_q(\mathbf{0}, \mathbf{J}_\infty^{-1})$, with the three statistics of Proposition 34.5.3 behaving as they do there.

We do not prove it; treat it as imported. Its first two steps are those of Theorem 34.5.1 — a Lyapunov limit theorem for the score, which [Proposition 36.2.2](#) exhibits as a sum of independent mean-zero terms, and a uniform quadratic expansion on compacts — but not its third, which used the global concavity of a canonical-link generalized linear model. For the classical treatment see van der Vaart (1998, chapter 5) or Lehmann and Casella (1998, chapter 6). Nothing here is exact in finite samples.

36.2.4 Where people place themselves

Example. A seven-point scale

The 1996 American National Election Study asked 944 respondents to place themselves on a seven-point scale from extremely liberal (1) to extremely conservative (7); the counts are 16, 103, 147, 256, 170, 218 and 34. With the respondent's age in decades, a seven-point education scale and a twenty-four-point household income scale as regressors, Fisher scoring for the common-slope cumulative logit model converges in 11 steps to

	age (decades)	education	income
$\hat{\beta}$	0.0942	-0.1249	0.0276
standard error	0.0357	0.0395	0.0104
z	2.637	-3.166	2.641

with cutpoints $\hat{\theta} = (-3.7744, -1.6378, -0.6159, 0.5580, 1.3625, 3.6442)$, in order as [Proposition 36.2.1](#) requires, and maximized log-likelihood -1613.8558. Older and richer respondents place themselves further to the right, more educated respondents further to the left: each decade of age multiplies the odds of a higher placement by 1.0988, each step up the education scale by 0.8826. Whether one should believe a single common slope is the question of [Section 36.3](#), and for these data the answer turns out to be no.

[Figure 36.2.1](#) shows what the fit says. Panel (a) plots the six fitted cumulative logits against age: exactly parallel lines, which is what a common slope means. Panel (b) differences them into the seven category probabilities, whose bands shift smoothly to the right as age increases and can never cross — the ordering constraint in picture form.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

NAMES = ["age (decades)", "education", "income"]

anes = sm.datasets.anes96.load_pandas().data
y = anes["selfLR"].astype(int).to_numpy()          # 1 = extremely liberal ... 7 =
    extremely conservative
X = np.column_stack([anes["age"] / 10.0, anes["educ"], anes["income"]])
n, p = X.shape
c = 7
Y = np.zeros((n, c))
Y[np.arange(n), y - 1] = 1.0

print("category counts:", Y.sum(axis=0))
```

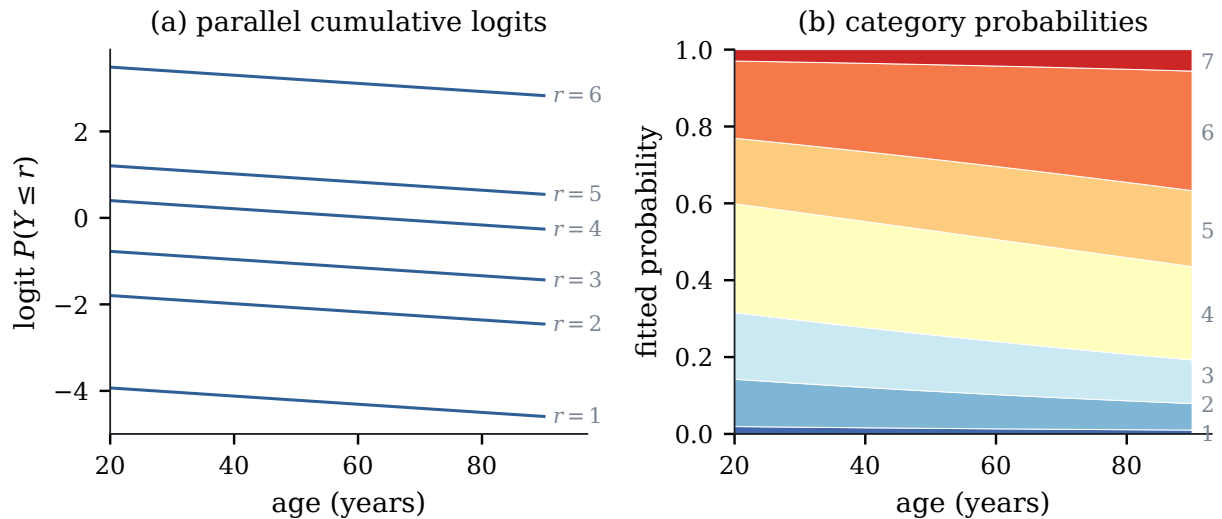


Figure 36.2.1: The fitted cumulative logit model for self-placement, as a function of age at education 4 and income 17. (a) The six cumulative logits are parallel, because the model gives them a common slope. (b) The same fit as category probabilities.

[Run this code online](#)

The fitting routine below is [Eq. 36.2.4](#) written out, with any subset of the regressors allowed cut-specific slopes through the argument `free`. Empty, it is the proportional odds fit; the other settings are for [Section 36.3](#).

```
def cum_gamma(psi, X, c, free):
    """Cumulative probabilities  $\Lambda(\theta_r - x^T \beta_r)$  and their gradients.

    psi = [theta (c-1), beta (p), delta ((c-2) x len(free))]; the columns of X
    listed
    in `free` get cut-specific slopes beta_r = beta + delta_r, with delta_1 = 0.
    """
    n, p = X.shape
    q, nf, fl = len(psi), len(free), list(free)
    theta, beta = psi[:c - 1], psi[c - 1:c - 1 + p]
    delta = psi[c - 1 + p:].reshape(c - 2, nf)
    G = np.empty((n, c - 1))
    dG = np.zeros((n, c - 1, q))
    for r in range(c - 1):
        beta_r = beta.copy()
        if r >= 1 and nf:
            beta_r[fl] = beta_r[fl] + delta[r - 1]
        g = 1.0 / (1.0 + np.exp(-(theta[r] - X @ beta_r)))
        lam = (g * (1.0 - g))[:, None]  # the logistic density at the
    cut
    G[:, r] = g
    dG[:, r, r] = lam[:, 0]
    dG[:, r, c - 1:c - 1 + p] = -lam * X
    if r >= 1 and nf:
        off = c - 1 + p + (r - 1) * nf
        dG[:, r, off:off + nf] = -lam * X[:, fl]
    return G, dG

def cum_probs(psi, X, c, free):
```

```

"""Category probabilities pi_ir and their derivatives D[i, r, k] = d pi_ir / d
psi_k."""
G, dG = cum_gamma(psi, X, c, free)
P = np.diff(np.column_stack([np.zeros(len(X)), G, np.ones(len(X))]), axis=1)
D = np.concatenate([dG[:, :1, :], np.diff(dG, axis=1)], axis=1)
return P, D

def cum_score(psi, X, Y, c, free):
    """Score and expected information of the multinomial likelihood."""
    P, D = cum_probs(psi, X, c, free)
    Pm = P[:, :c - 1]
    Sigma = np.einsum("ir,rs->irs", Pm, np.eye(c - 1)) - np.einsum("ir,is->irs", Pm,
        Pm)
    A = np.linalg.solve(Sigma, D) # Sigma_i^{-1} D_i, one
    observation at a time
    U = np.einsum("irk,ir->k", A, Y[:, :c - 1] - Pm)
    J = np.einsum("irk,irl->kl", D, A)
    return U, J

def cum_loglik(psi, X, Y, c, free):
    P, _ = cum_probs(psi, X, c, free)
    return float(np.sum(Y * np.log(np.clip(P, 1e-300, None))))

def cum_fit(X, Y, c, free=(), tol=1e-10, maxit=100):
    """Fisher scoring for the cumulative logit model."""
    n, p = X.shape
    psi = np.zeros((c - 1) + p + (c - 2) * len(free))
    psi[:c - 1] = np.linspace(-1.5, 1.5, c - 1) # start from ordered cutpoints,
    no slopes
    for it in range(1, maxit + 1):
        U, J = cum_score(psi, X, Y, c, free)
        step = np.linalg.solve(J, U)
        psi = psi + step
        if np.max(np.abs(step)) < tol:
            break
    return psi, J, it

psi, J, iters = cum_fit(X, Y, c)
theta, beta = psi[:c - 1], psi[c - 1:]
se = np.sqrt(np.diag(np.linalg.inv(J)))
ll_po = cum_loglik(psi, X, Y, c, ())

print("cutpoints", np.round(theta, 4))
for j, name in enumerate(NAMES):
    print(f"{name:>16s} {beta[j]:8.4f} ({se[c - 1 + j]:.4f}) z = {beta[j] / se[c - 1 + j]:6.3f}")
print(f"log-likelihood {ll_po:.4f} after {iters} Fisher scoring steps")

```

[Run this code online](#)

The script checks the fit, cutpoint by cutpoint, against the `OrderedModel` routine of `statsmodels`.

36.2.5 Choosing the link

The three links of [Definition 36.2.1](#) are close over the middle of the probability range and differ in the tails, as in the binary case ([Proposition 35.2.1](#)). Two things are specific to ordinal data. First, the *direction* of the scale is arbitrary. Relabelling the categories in reverse order maps the logit and probit models to themselves with β and the cutpoints negated ([Exercise \(A1\)](#)), but turns the complementary log-log model into a different one, the log-log model; with an asymmetric link the analysis depends on which end of the scale was called “1”, and the choice must be justified.

Second, the complementary log-log link has an interpretation the others lack. With $G(z) = 1 - \exp(-e^z)$ and common slopes, [Eq. 36.2.2](#) gives

$$P(Y_i > r \mid \mathbf{x}_{(i)}) = \exp\{-e^{\theta_r} e^{-\mathbf{x}_{(i)}^\top \beta}\} = [P(Y > r \mid \mathbf{0})]^{\exp(-\mathbf{x}_{(i)}^\top \beta)}, \quad (36.2.5)$$

so the regressors act multiplicatively on the cumulative hazard of the underlying scale: the grouped-data version of the proportional hazards model of Cox (1972). If the ordered categories are intervals of a survival time, reach for it.

36.2.6 Exercises

36.2.6.1 A. Check your understanding

Exercise. A1

Verify the claim that reversing the category order maps the cumulative logit model to itself with β and the cutpoints negated, and find what it does to the complementary log-log model.

Solution. Let $Y' = c + 1 - Y$. Then $\{Y' \leq r\} = \{Y \geq c + 1 - r\} = \{Y > c - r\}$, so $\gamma'_{ir} = 1 - \gamma_{i,c-r}$. Since $\Lambda^{-1}(1 - u) = -\Lambda^{-1}(u)$, $\Lambda^{-1}(\gamma'_{ir}) = -\theta_{c-r} + \mathbf{x}^\top \beta$, which is [Eq. 36.2.2](#) with cutpoints $-\theta_{c-r}$ (still increasing in r) and slope $-\beta$. The same works for Φ . For $G(z) = 1 - \exp(-e^z)$ the identity $G^{-1}(1 - u) = -G^{-1}(u)$ fails, and $G^{-1}(1 - u) = \log(-\log u)$ defines the log-log link, a different family.

36.2.6.2 B. Practice

Exercise. B1

Show that in the proportional odds model the maximum likelihood estimate automatically satisfies $\hat{\theta}_1 < \dots < \hat{\theta}_{c-1}$, provided every category is observed, so the ordering constraint never binds. (Hint: what is the likelihood if two cutpoints coincide or cross?)

Solution. The likelihood is defined only where [Eq. 36.2.3](#) holds, and $\pi_{ir} \rightarrow 0$ for some i and r as the boundary $\theta_r = \theta_{r+1}$ is approached. If category r is observed at least once, some $y_{ir} > 0$ and $\ell \rightarrow -\infty$ there. Since ℓ is continuous on the open region and tends to $-\infty$ at the part of its boundary where two cutpoints meet, any maximizer lies in the interior, where the cutpoints are strictly ordered.

Exercise. B2

The cumulative logit model is unchanged if the latent scale is multiplied by a positive constant a and shifted by b , provided $\theta_r \mapsto a\theta_r + b$ and $\boldsymbol{\beta} \mapsto a\boldsymbol{\beta}$. Deduce that only *ratios* of coefficients, and the standardized distances between cutpoints, are free of the arbitrary scale — and that a coefficient from a logit fit is not comparable with a coefficient from a probit fit without rescaling by roughly 1.8, the standard deviation of the logistic distribution.

Exercise. B3

Refit [Example \(A seven-point scale\)](#) after collapsing the seven categories into three (1–3, 4, 5–7) and again after dichotomizing at the middle category. Compare the three sets of slopes and their standard errors, and say which comparison is a check on the model and which is a statement about information.

Solution. The estimates from the three fits are $(0.0942, -0.1249, 0.0276)$, $(0.1350, -0.1148, 0.0343)$ and $(0.1229, -0.0311, 0.0398)$, with standard errors that grow as categories are merged. That they estimate the *same* parameters is the collapsing invariance proved in [Section 36.3](#) — a check on the model, since it holds only if the proportional odds model is true. That the standard errors grow is a statement about information: collapsing throws data away.

36.2.6.3 C. Going deeper**Exercise. C1**

Suppose an underlying continuous response U_i follows the linear model $U_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \varepsilon_i$ with ε_i having distribution function G , but U_i is recorded only as the interval $(\theta_{r-1}, \theta_r]$ it falls in, with *known* cutpoints. Write down the log-likelihood, show it is the cumulative link model with the cutpoints held fixed, and say what is gained by knowing them ([Exercise \(B2\)](#)).